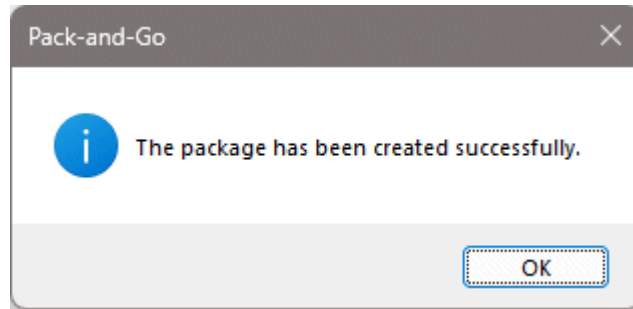




- The recipient can double-click the file you sent and start the simulation to view what you wanted to show.

As the *Pack-and-Go* model uses the **Plant Simulation Viewer License** only a limited set of functions is available.

Visualizing the Material Flow

Visualizing the Material Flow

Introduces you to the most important tasks you will face when you create and visualize your simulation model.

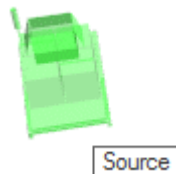
Plant Simulation differentiates between these object types.

- **Simulation Object**
- **Animatable Object**
- **Graphic Group [defined]**
- **State Graphic [defined]**
- **Graphic [defined]**

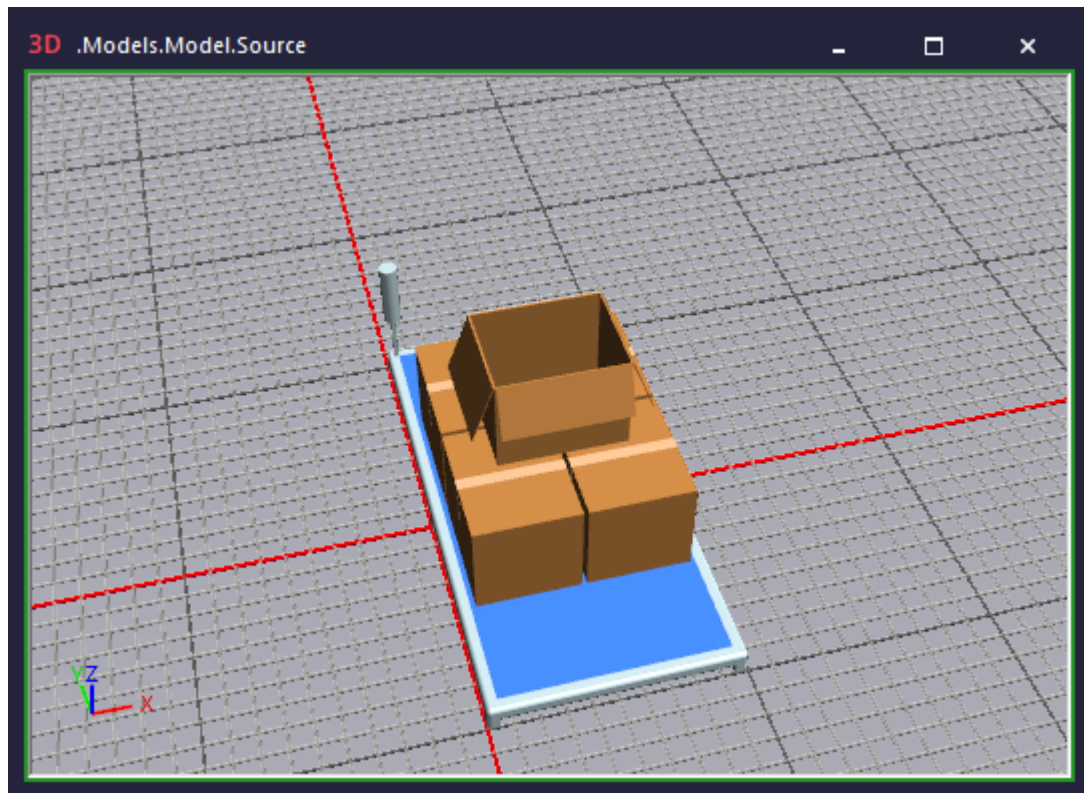
The dialog **Show Graphic Structure [context menu]** show these object types in a tree structure.

A 3D window shows the objects with these colors:

- *Plant Simulation* shows a selected **simulation object** in **green** in the 3D window.



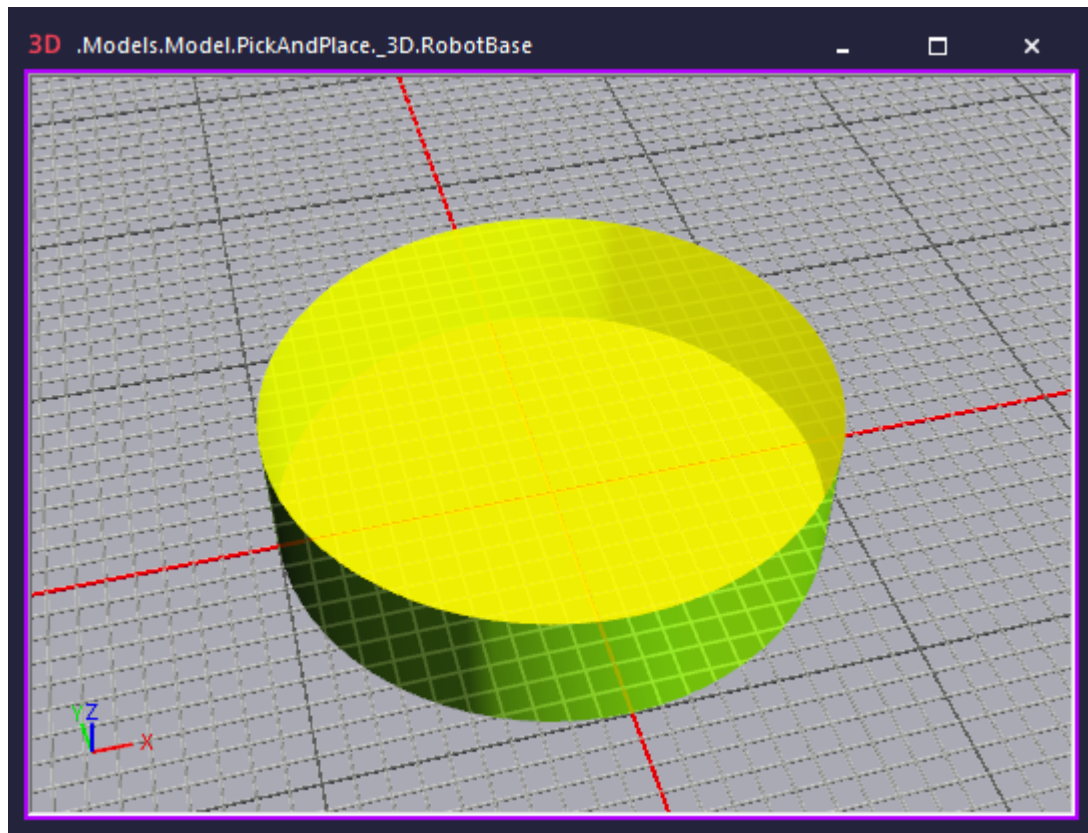
When you open a **simulation object**, the 3D window shows a **green border** on the inside.



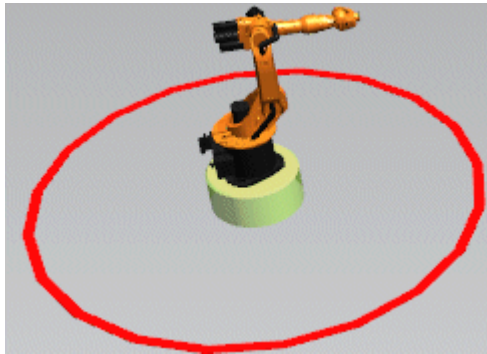
- *Plant Simulation* shows a selected **animatable object** in **purple** in the 3D window.



When you open an **animatable object**, the 3D window shows a **purple border** on the inside.



- *Plant Simulation* shows a selected **graphic** in **yellow green** in the 3D window.



Some of the videos on the **Start Page** provide an introduction to 3D or introduce certain modeling tasks in 3D.

Compare [3D Properties Visualizing the Simulation](#)

Compare [SimTalk Access to 3D Functions](#)

Compare [Modeling the Material Flow](#)

Compare [Animate the Simulation Model and View the Results](#)

Compare [Getting to Know Plant Simulation](#)

Creating a Simulation Model

You create your simulation model in several steps.

- To create a new *Plant Simulation* model, click **Create New Model** on the **Start Page** or select **File > New**.
- To show the built-in simulation objects and their graphics, you can, for example, click the folder **MaterialFlow** in the **Class Library** with the right mouse button and select **Open in 3D**. This way you can also show the contents of the other folders.

To show the name of an object, drag the mouse over it.

To **rotate** the scene, hold down the left and the right mouse buttons and drag the mouse or hold down **Ctrl** and the right mouse button and drag the mouse. The rotate mouse pointer looks like this .

To **pan** the scene, hold down the right mouse button and drag the mouse. The pan mouse pointer looks like this .

To **move** the camera forward and backward, roll the mouse wheel.

To **zoom** the scene in or out, roll the mouse wheel or hold down **Shift** and the right mouse button and drag the mouse.

Note

If you have a three-button mouse, click the middle mouse button where the description calls for clicking the mouse wheel.

Note

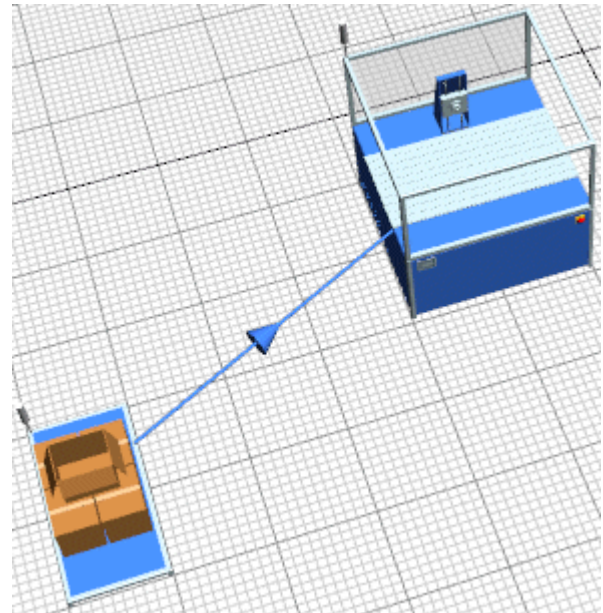
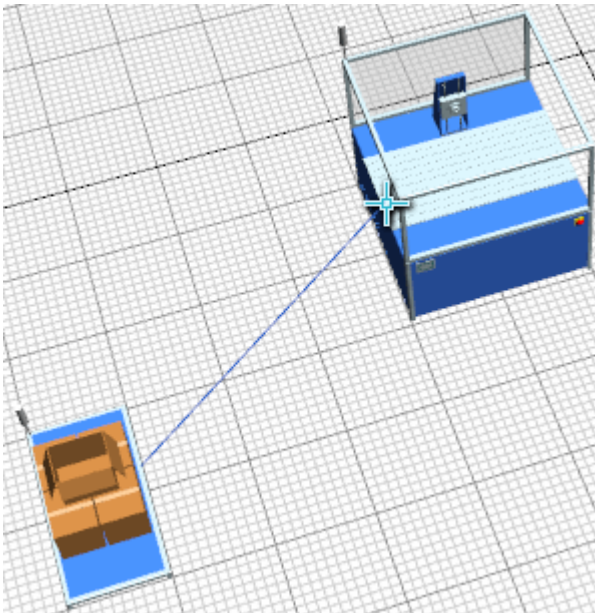
If manipulating the scene with the mouse does not work, check the **Mouse Properties** for the **Wheel** in the **Mouse Control Panel** under **Start > Control Panel**.

- To open a new window, change to the **Class Library**. Then, click the corresponding folder with the right mouse button and select **Open in 3D**.
- To insert an object into the window:
 - Make sure that **Show Grid**  on the **View** ribbon tab is active, compare [Show and Hide the Grid](#).
We recommend to show the grid before you insert an object into the scene window. This way you can place the object at the exact position, where you want it to be.
If you insert an object without displaying the grid, you have little control over where *Plant Simulation* actually places the object. By default, it uses the same coordinates that the source object had, an effect you might not want.
 - Select the object you want to insert in the *Toolbox*. Drag the mouse pointer to the position of your choice and click the left mouse button to insert it there.

As long as an object is selected in the *Toolbox*, *Plant Simulation* shows a preview of the object in the window to make it easier for you to position it.

To cancel inserting objects, click the right mouse button.

- Insert a *Source*, a *Conveyor*, a *Station* and a *Drain*.
- Connect the inserted simulation objects:
 - Make sure that **Show Connections**  on the **View** ribbon tab is activated as 3D does not show the *Connectors* by default.
 - Click the *Connector*  in the *Toolbox* on the tab **MaterialFlow**.
 - Click the source object, where the connection will start, once with the left mouse button. *Plant Simulation* attaches a line symbolizing the *Connector* to the mouse pointer.



- Drag the mouse to the destination object at which the connection will end, and click once with the left mouse button to establish the connection.

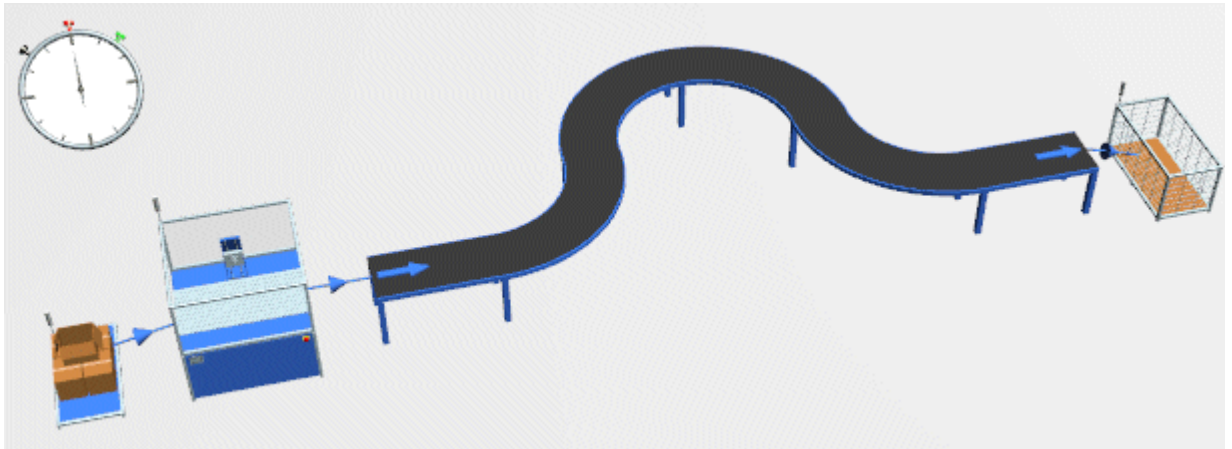
3D shows the *Connector* it inserted as a line between the connected internal interfaces of the source and destination objects. The pointed cone shows the direction of the *Connector*.

- To terminate connect mode before you click the destination object, click the right mouse button or press **Esc**.

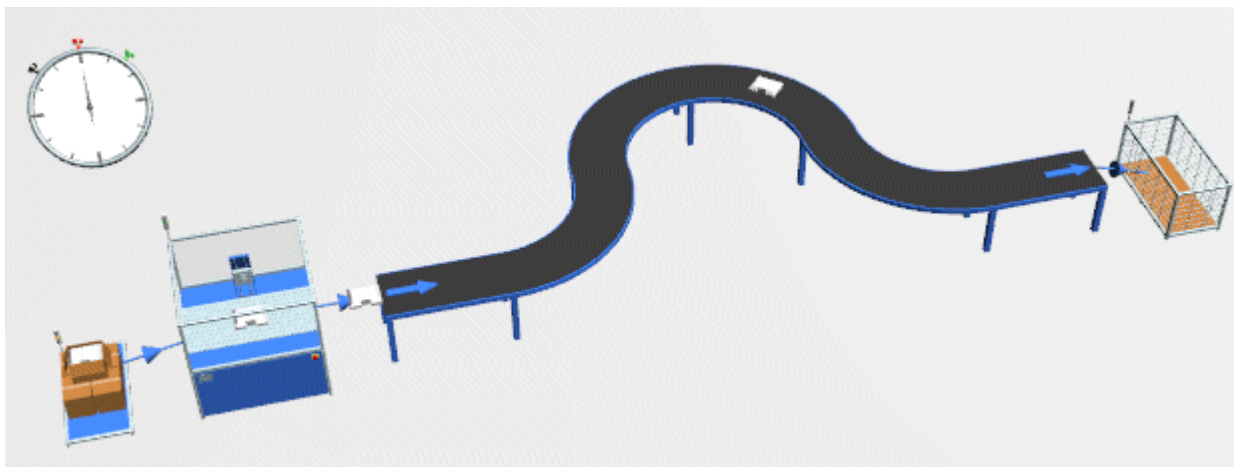
To connect several objects one after the other, without interrupting the connecting process, hold down **Ctrl**.

- ♦ Attach the outgoing connection to the first object.
- ♦ Drag the mouse to the succeeding object.
- ♦ Click the left mouse button to attach the incoming *Connector* to the successor.

- ◆ Connect the *Source*, the *Conveyor*, the *Station* and the *Drain* you inserted above.



- Start the simulation:
 - ◆ Control the simulation by clicking the buttons on the **Home** ribbon tab or
 - ◆ Double-click the icon of the *EventController*  and click its buttons.
 - ◆ Watch how the produced parts move across the simulation objects.



Compare [Work with Objects](#)

Compare [Work with the Grid](#)

Compare [Control Your View](#)

Compare [Work with the Scene](#)

Back to [Visualizing the Material Flow](#)

Modeling Hierarchically [in 3D]

Using hierarchical modeling techniques you can add any level of detail to the areas and machines you insert into your simulation model. Hierarchical modeling means that you create a machine, a production area, etc. in a *Frame* using the built-in object classes and/or classes you yourself designed.

You can then insert this *Frame*, *Machine* for example, into another *Frame* in which you modeled a production area, and then insert the production area into the *Frame* that contains your entire factory.

Compare [Model Hierarchically \[general\]](#)

Compare [Show Reports in a Hierarchically Structured Model](#)

Back to [Visualizing the Material Flow](#)

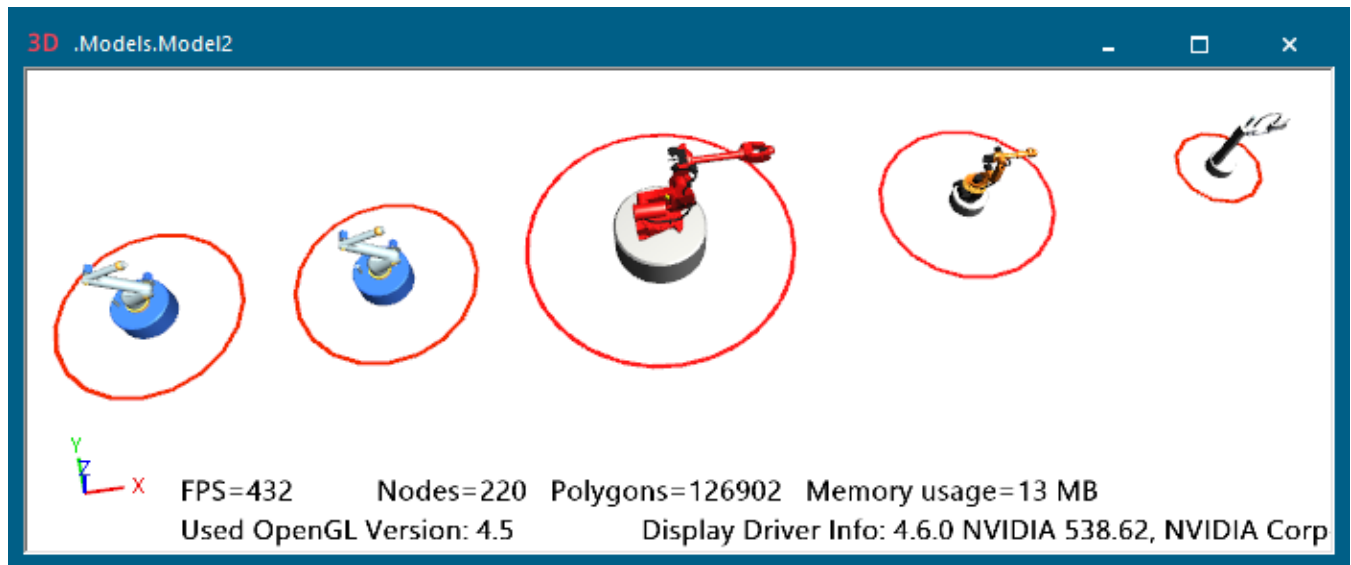
Working with the Scene

Work with the Scene

Plant Simulation provides a number of ways for working with the active scene, as opposed to working with the objects you insert into that scene.

You can:

- [Manipulate the Scene with the Mouse](#)
- [Align the View to the Main Directions](#)
- [Save a View with the Model and Return to It](#)
- [Fly on a Defined Path Through the Scene](#)
- Show information about the open scene by pressing the **F** key on the keyboard. To hide the information, press the **F** key again. The scene window shows:
 - The **Frames per Second** (FPS).
 - The summed up **Number of Nodes** of the scene window.
 - The summed up **Number of Polygons** of the scene window.
 - The summed up **Memory usage** of the scene window.
 - The [OpenGL Version](#) used.
 - Information about the **Display Driver** of your computer.



Compare [Graphic Complexity](#)

Compare [Work with Objects](#)

Compare [Control Your View](#)

Compare [Work with the Grid](#)

Back to [Visualizing the Material Flow](#)

Manipulate the Scene with the Mouse

You can manipulate the scene itself and the objects within that scene with the mouse. You can move within the scene and change your view of the scene.

Instead of using a normal mouse, you can also navigate in the scene with a Space Navigator mouse.

Note

If you have a three-button mouse, click the middle mouse button where the description calls for clicking the mouse wheel.

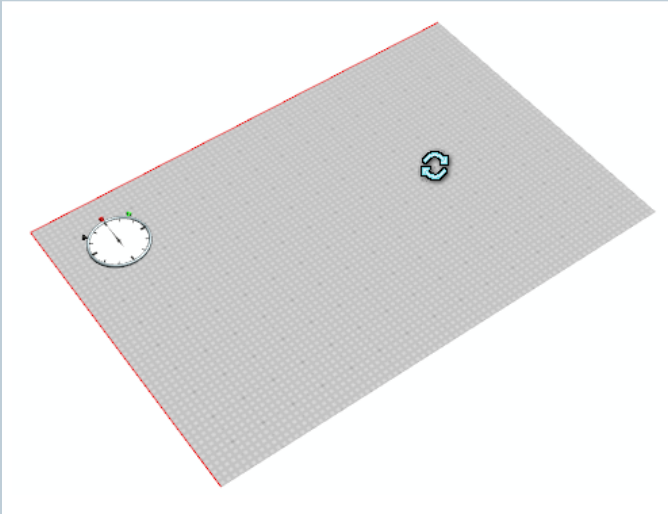
If manipulating the scene with the mouse does not work, check the **Mouse Properties** for the **Wheel** in the **Mouse Control Panel** under **Start > Control Panel**.

- To **rotate the scene**, hold down the left and the right mouse buttons and drag the mouse, or hold down **Ctrl** and the right mouse button and drag the mouse.

Mouse pointer, rotate around center of the scene	Mouse pointer, rotate around object
	

When you start rotating a view with the mouse and remain at the position for at least 400ms and if an object or a graphic is located under the mouse pointer, *Plant Simulation* rotates around the position under the mouse pointer on the object or the graphic.

In all other cases *Plant Simulation* rotates around the projection of the center of the window on the insertion plane, or around a matching orthogonal of this plane if the view is rotated by 90°. Here it does not matter where the mouse pointer is positioned.

Rotate around center of the scene	Rotate around object
	

- To **pan the scene**, hold down the right mouse button and drag the mouse. The pan mouse pointer looks like this . If you have a three-button mouse, you can also hold down the middle mouse key and drag the mouse.
- To **move the camera** forward and backward, roll the mouse wheel, or hold down **Shift** and the right mouse button and drag the mouse.

To move the scene with a **tenth of the speed**, hold down **Ctrl** and roll the mouse wheel.

To move the scene with **ten times the speed**, hold down **Shift** and roll the mouse wheel.

In **Planning View** you can navigate like this:

- To **move the scene on the plane**, hold down the right mouse button and drag the mouse.
- To **zoom the scene**, roll the mouse wheel, or hold down **Shift** and the right mouse button and drag the mouse.

To zoom the scene with a **tenth of the speed**, hold down **Ctrl** and roll the mouse wheel.

To zoom the scene with **ten times the speed**, hold down **Shift** and roll the mouse wheel.

Compare [Mouse Wheel Behavior](#)

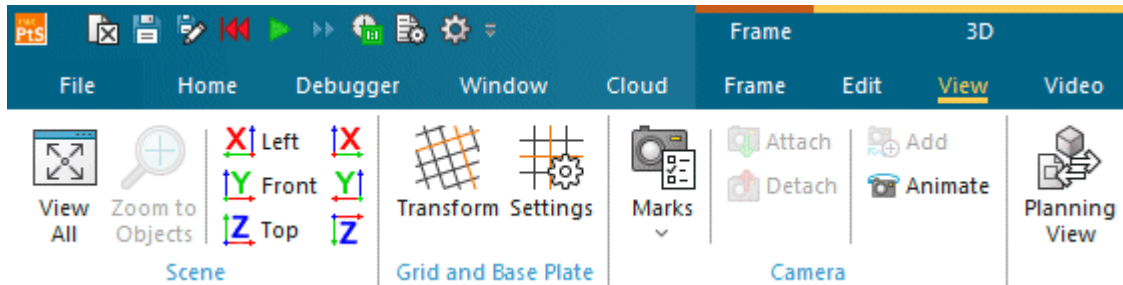
Compare [Move an Object With the Mouse](#)

Back to [Work with the Scene](#)

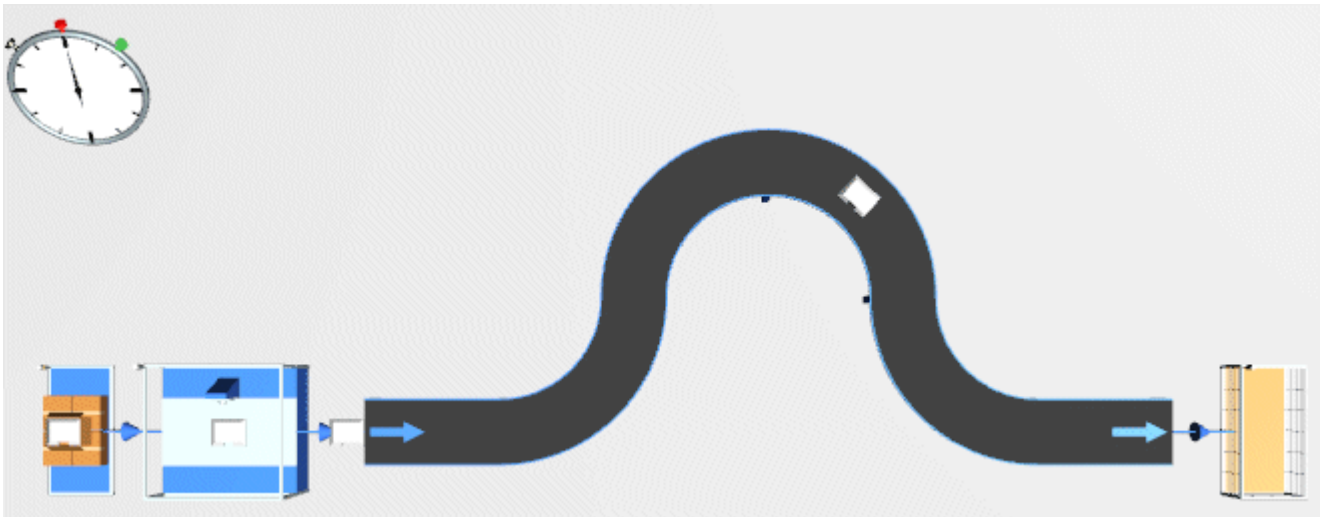
Align the View to the Main Directions

You can select from which predefined direction you look at the scene using the commands on the **View** ribbon tab.

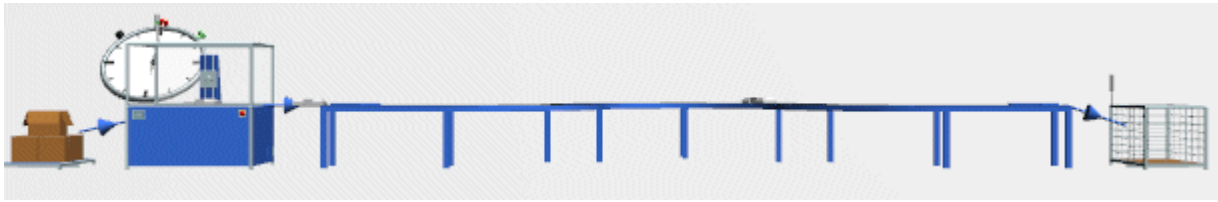
From the **Right**, from the **Left**, from the **Back**, from the **Front**, from the **Top**, or from the **Bottom**.



- To view the entire scene from the **top** looking down, click  on the **View** ribbon tab.

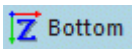


- To view the entire scene from the **front**, click .



- To view the entire scene from the **left side**, click  **Left**.



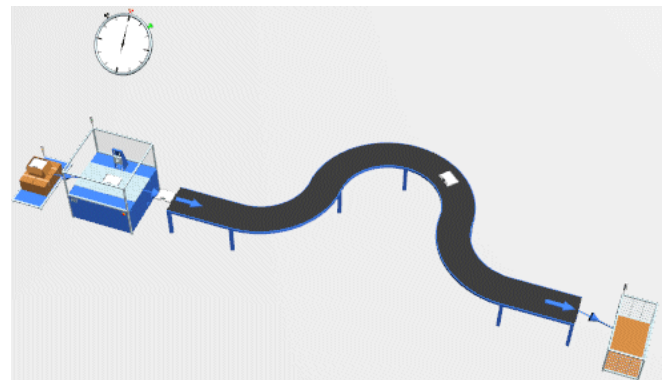
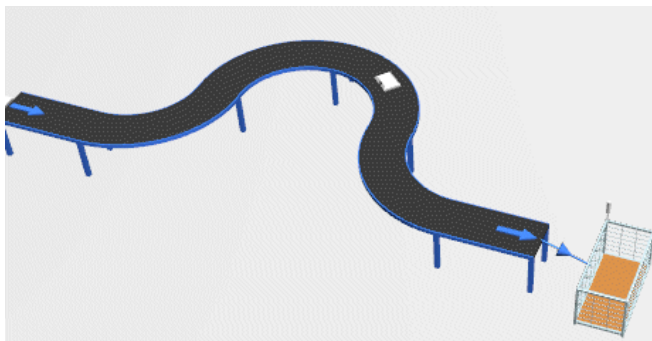
- To view the entire scene from the **bottom**, click  **Bottom**. This shows the entire scene from the **bottom** looking upward, i. e., along the negative z-axis.

If **Show Grid** is activated, *Plant Simulation* shows the grid below the object that you want to see from below.

If **Show Base Plate** is activated, *Plant Simulation* shows the base plate below the object that you want to see from below. The base plate then hides the object as such.

If you selected anything in the scene, the command adjusts the scene so that the selection is visible instead of the entire scene.

- To fit the entire scene into the scene window and to show all objects in the scene if you moved part of the scene out of view, click **View All** .



Back to [Work with the Scene](#)

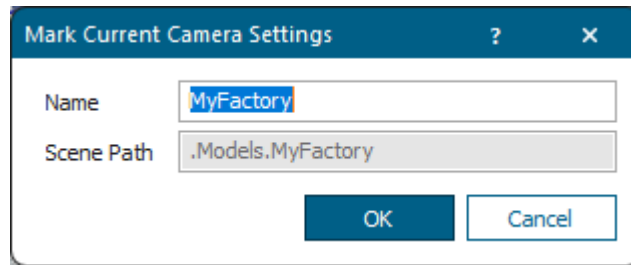
Save a View with the Model and Return to It

You can save a view of the simulation model with the model and return to it later on.

To save the current position and orientation of the camera with the model file, click **Camera Marks** on the **View** ribbon tab in 3D.



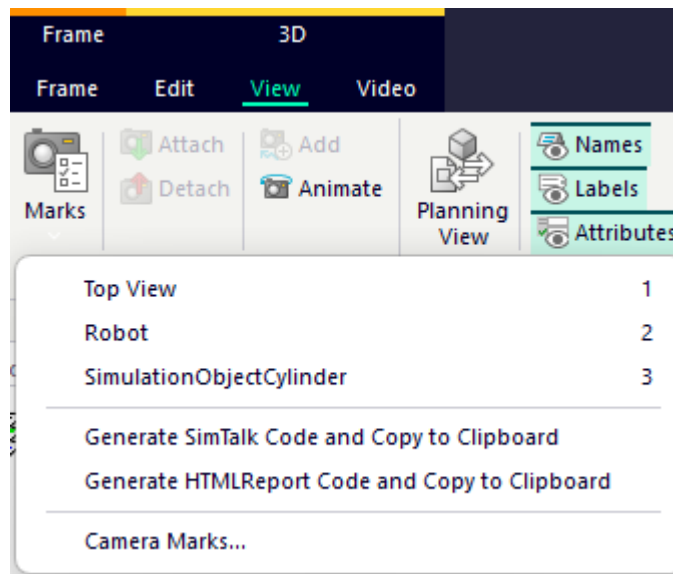
- Enter a **Name** for the current camera setting into the dialog **Mark Current Camera Settings**.



- 3D shows the path of the scene in the field **Scene Path**.
- Click **OK** to set the name. This saves this setting together with the active root object of the scene in the model file. Then, you can easily return to this saved view after you changed the scene or after you reopened the model.

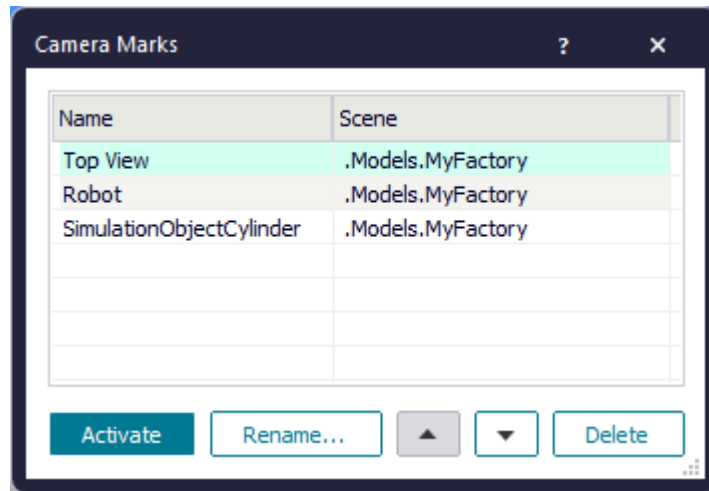
To return to a view you saved later on, or to navigate to the different predefined folders:

- Click the down arrow on the **Camera Marks** **Marks** button. Select the camera mark, i.e., the view for which you defined it, on the list.



Or

- Select the command **Camera Marks**.
- Select one of the saved scenes in the list and click **Activate**.



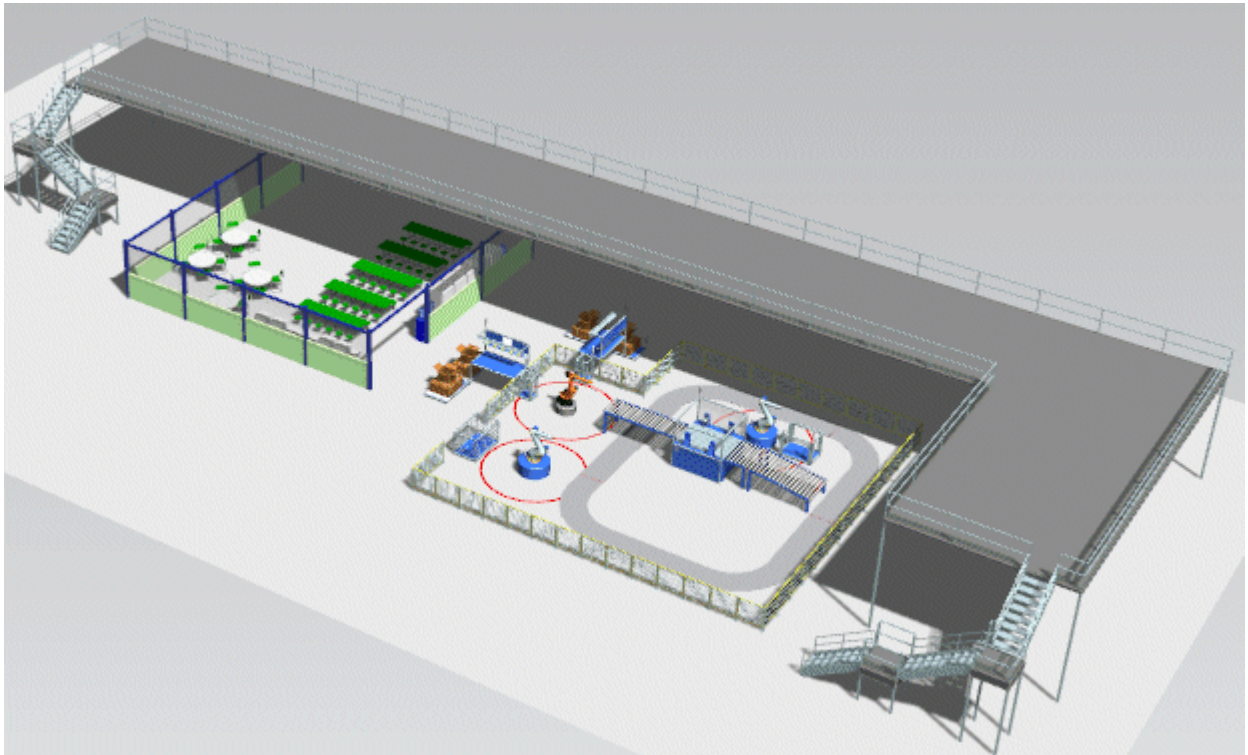
To rename or delete a camera mark, select it in the list, and click **Rename** or **Delete**.

To move the selected camera mark up in the list, click .

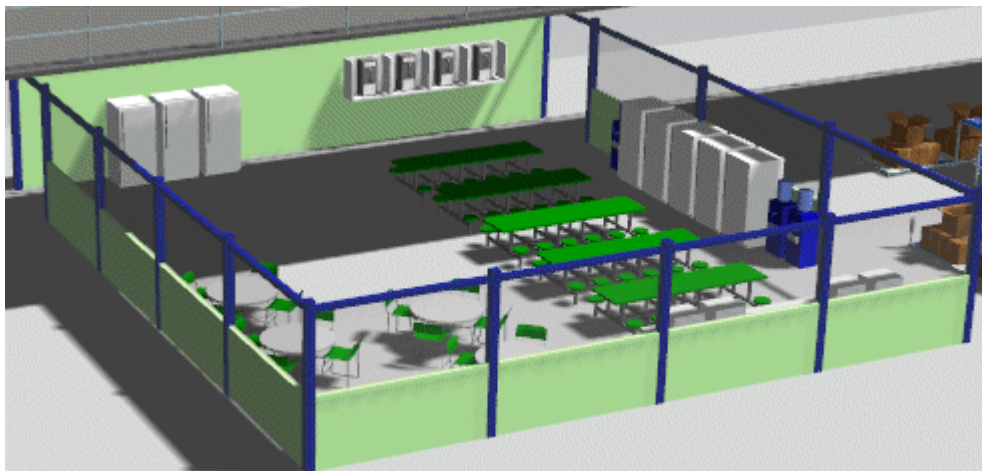
To move the selected camera mark down in the list, click .

Compare these examples:

- The **total view** of our model looks like this:



- The **cafeteria** of our model looks like this:



Back to [Work with the Scene](#)

Set the Background Color of the Scene

You can set the **Background Color** of the selected *Frame* or the selected folder in the *Class Library* in the 3D window.

Proceed as follows:

- Click **Edit 3D Properties** on the **Home** ribbon tab or click into the *Frame* window with the right mouse button and select **Edit 3D Properties**. Then, click the **Tab Background**.

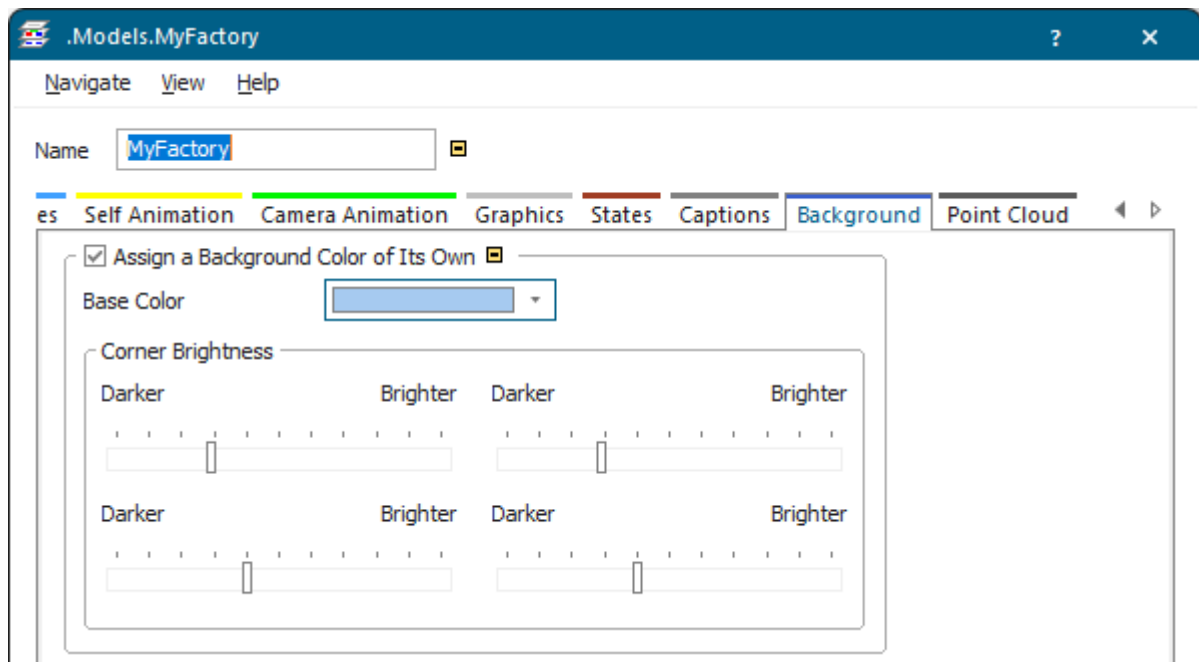
Note

You can also open the dialog **Edit 3D Properties** for the selected object by clicking the **spacebar**.

- Select **Assign a Background Color of Its Own**. Once you select the check box, *3D* activates the other settings on the tab.
- Select the **Base Color**. This is the base color of the background which you can change with the corner brightness settings.

You can select one of the predefined colors or you can click **More Colors** and click the **Select** button to select a color in the color matrix. Then click **OK**. *Plant Simulation* shows this color next to **More Colors** and uses it as the active color.

- Select settings for the **Corner Brightness** of the four corners of the *Frame* window. The top left slider sets the settings of the top left corner of the *Frame* window, etc. Drag the sliders to make the respective corner darker or brighter. This way you can define a color gradient, i.e., the brightness of the background color.



Note

If you do not define a background color, *Plant Simulation* uses the background color of the parent object of the selected object.

Back to [Work with the Scene](#)

Flying on a Defined Path Through the Scene

Fly on a Defined Path Through the Scene

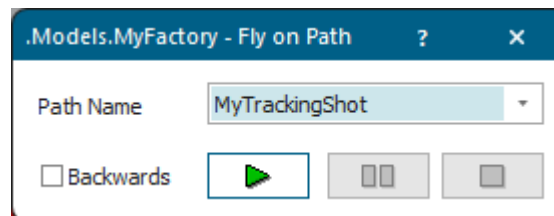
You can attach a camera to an object and view the scene *through the lens* of an object. This camera then moves along with the object through the scene during a simulation run. By turning or tilting the camera, you can control the direction and the angle at which you look at the scene.

3D visualizes the camera with a tool object as the icon of the camera. At present 3D provides three different cameras:

- The **main camera** through which you view the scene in normal view.
- The **object camera**, which you attach to a specific object and which moves with this object through the model. You might, for example, attach a camera to a part or a *Worker* to follow it or him on their way through the factory.
- The **animation camera**, which moves along an animation path you defined.

Proceed as follows to define a tracking shot through the scene:

- Click an object with the left mouse button to select it.
- Click  **Animate Camera** on the **View** ribbon tab to open the dialog **Fly on Path**. If you did not select an object before, the dialog refers to the current root object of the scene.
- Select one of the camera paths you defined to move your view point along this path.



You can also define your own camera paths or edit any existing path in the dialog **Edit 3D Properties > Tab Camera Animation** of the *Frame* to meet your specific modeling needs. Compare [How the Camera is Animated in the Frame](#).

You can:

- [Attach a Camera to an Object](#)
- [Detach the Camera from an Object](#)
- [Animate the Object Camera](#)

Compare [Ballhaus Kreisel](#)

Compare [Work with Animation Paths](#)

Compare [Tab Camera Animation](#) > [How the Camera is Animated in the Frame](#)

Back to [Work with the Scene](#)

Attach a Camera to an Object

You can attach a camera to an object, to view the scene through the lens of this camera.

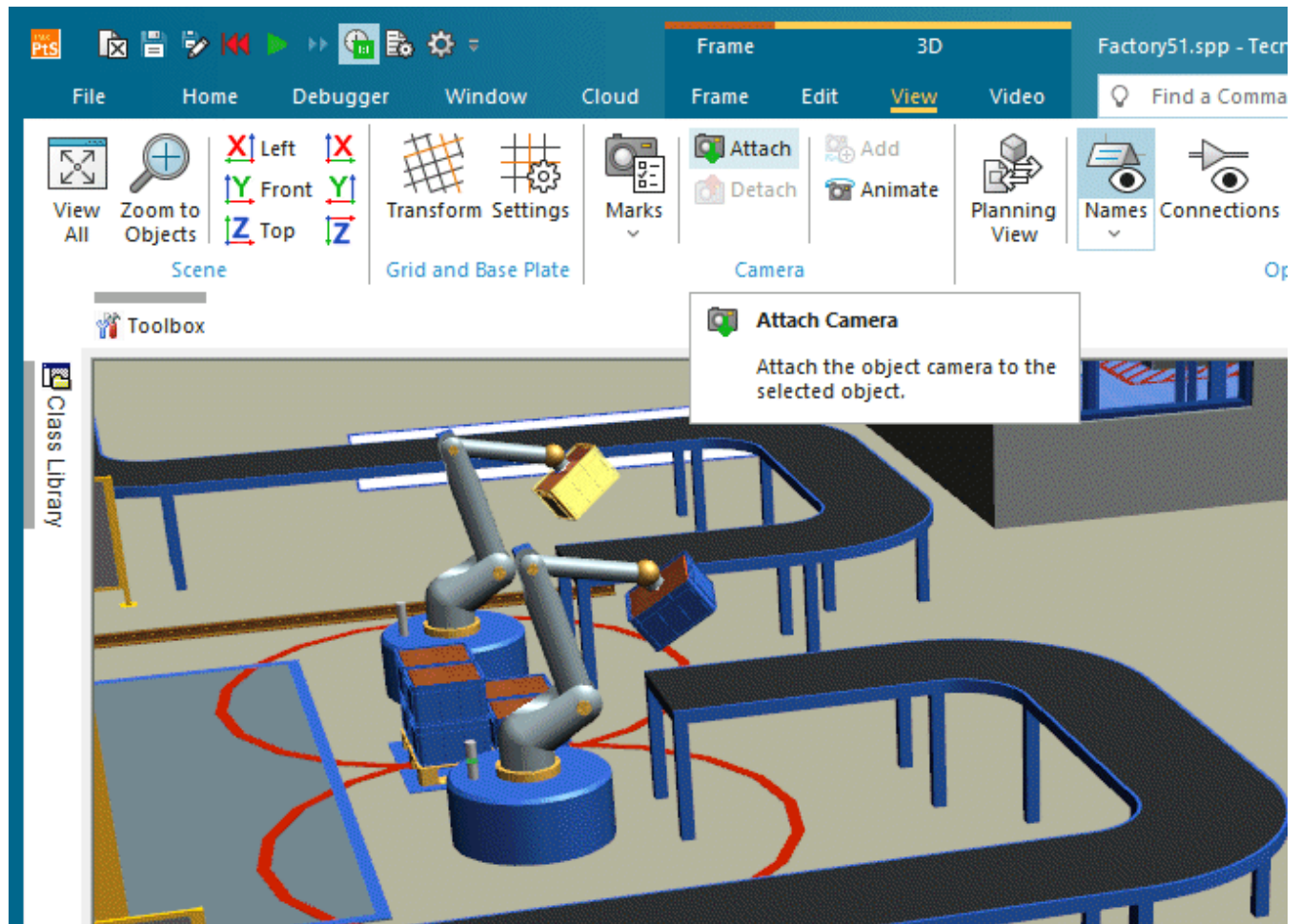
Proceed as follows:

- Select a single object.
- Click  [Attach Camera](#) on the **View** ribbon tab of the *3D window*.
- *Plant Simulation* attaches the camera at a defined position:

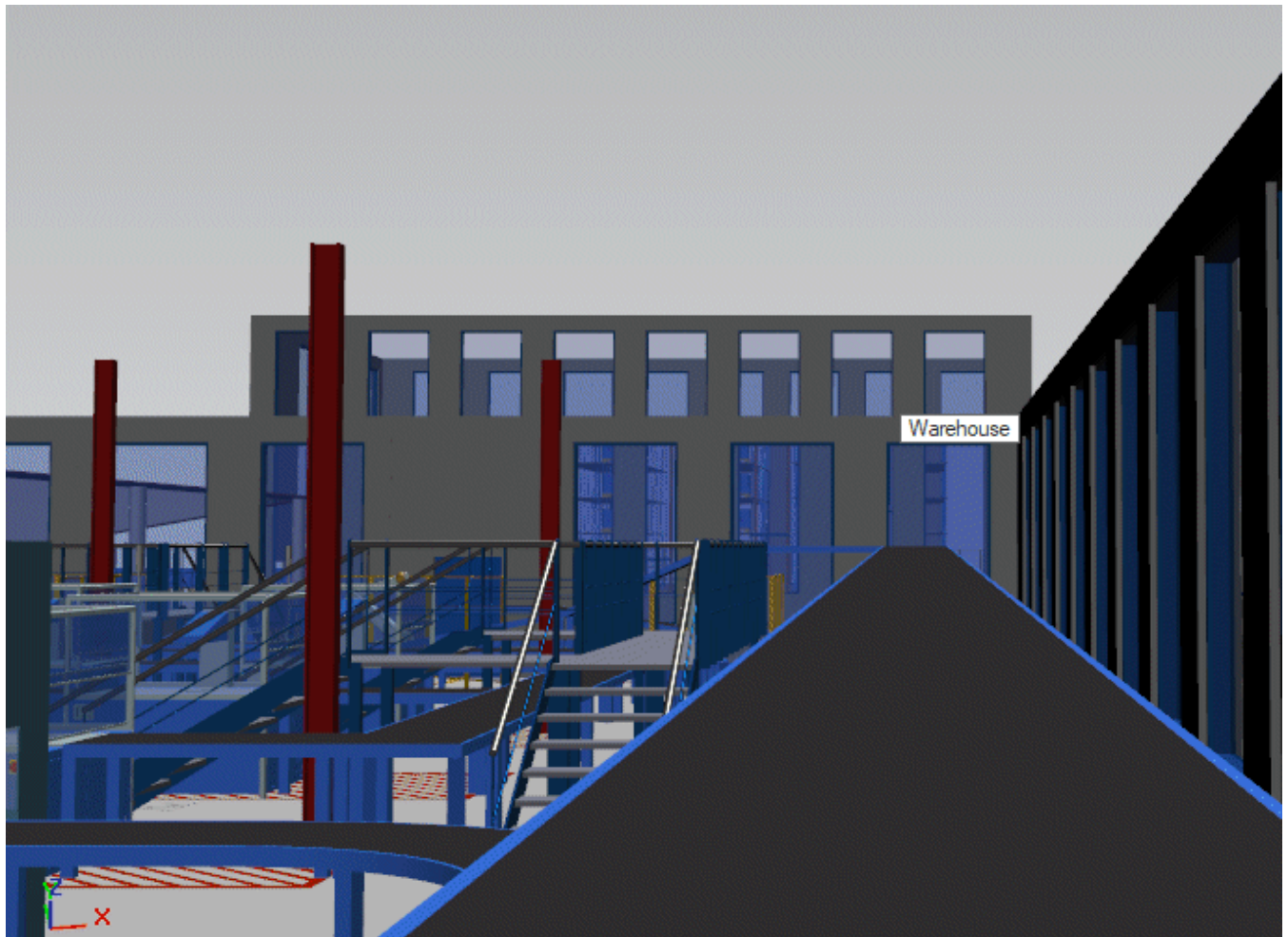
Plant Simulation uses the center of the top of the bounding box of the selected object as the position where it inserts the camera. *Plant Simulation* adds an offset of 0,0,0.1 to these makeshift points, so that your eyes will be located above the top plane.

By default the object camera looks along the positive x-axis using the positive z-axis as **Up direction**.

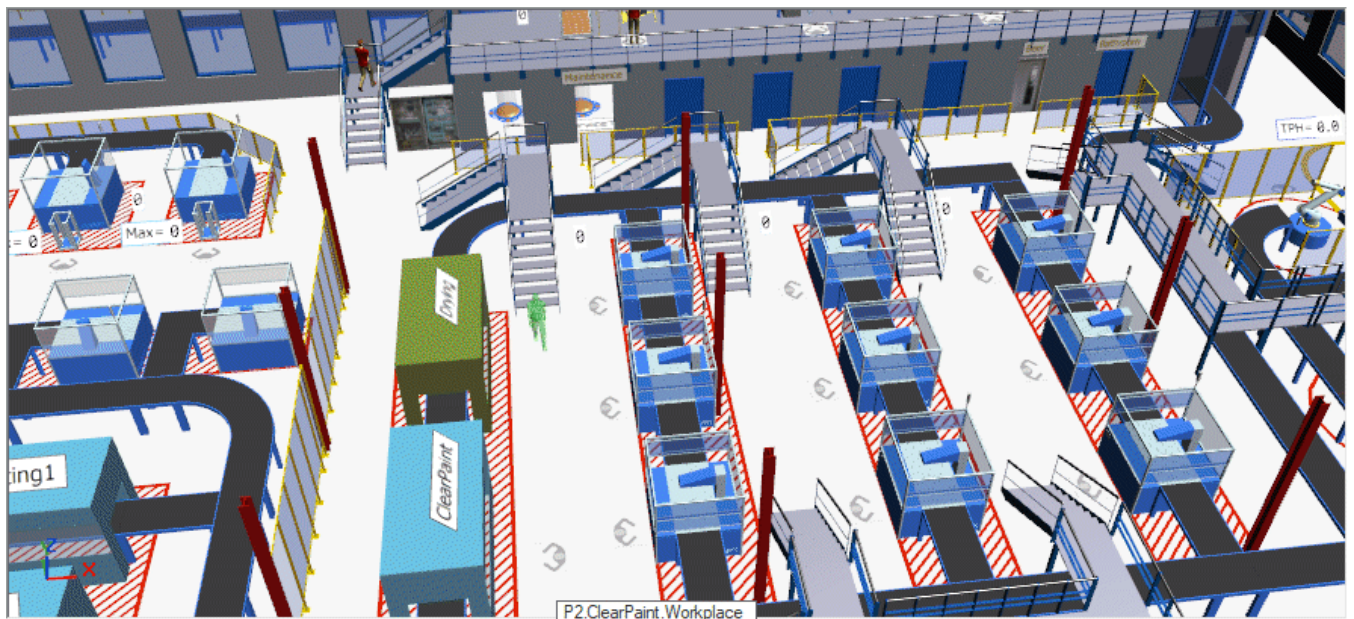
You might, for example, attach a camera to one of the boxes in our sample model **Factory51** and track its way through the factory.



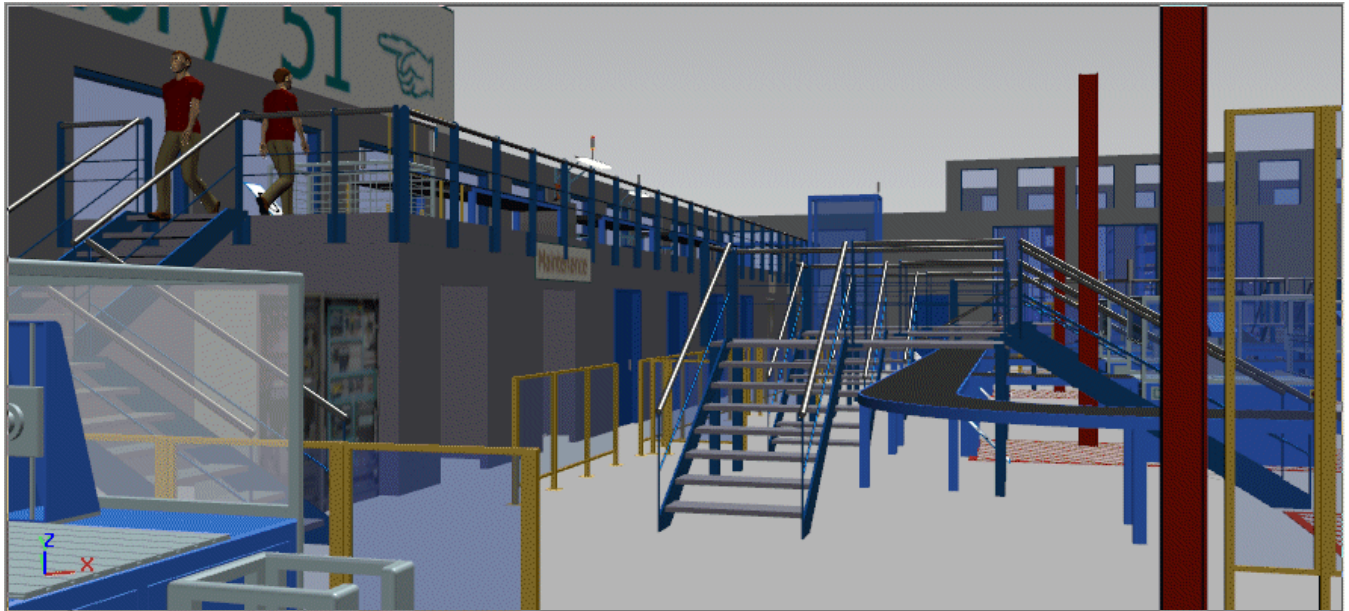
As soon as the object, with a camera attached to it, moves through the scene, the camera moves along with it automatically. Once the camera is attached, you see the scene through the lens of this object camera.



Or you can attach a camera to one of the *Workers* and track his way between the work stations.



You then see what the *Worker* sees while he walks and what he sees while he stands on the *Workplaces* and works at the machines.



Compare [How the Camera is Animated in the Frame](#)

Compare [Move an Object Precisely](#)

Compare [Work with Animation Paths](#)

Back to [Fly on a Defined Path Through the Scene](#)

Detach the Camera from an Object

If you do not need the camera, which you attached to an object, any longer, you can detach it from the object.

To detach the camera, click  **Detach Camera** on the **View** ribbon tab of the *3D window*. *Plant Simulation* then returns to normal view.

Plant Simulation also detaches an object camera:

- When you delete the object to which the camera is attached.
- When you change the scene.

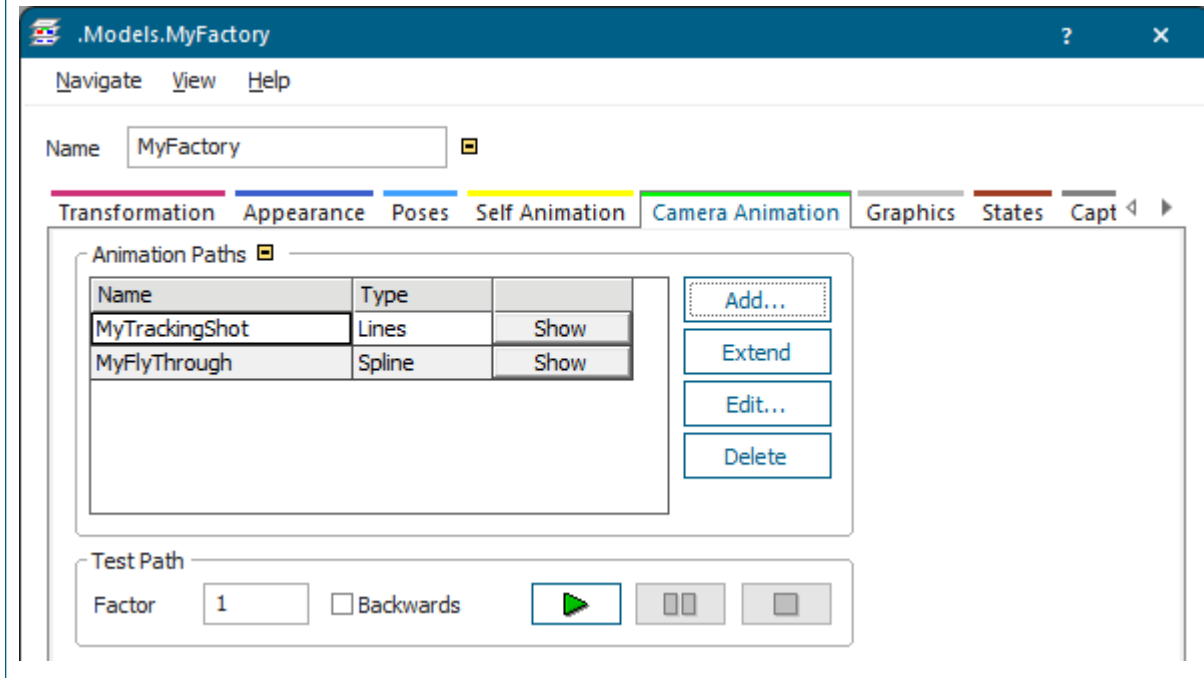
Back to [Fly on a Defined Path Through the Scene](#)

Animate the Object Camera

You can animate the camera on a camera animation path of the selected *Frame*, thus creating the impression of flying through the scene.

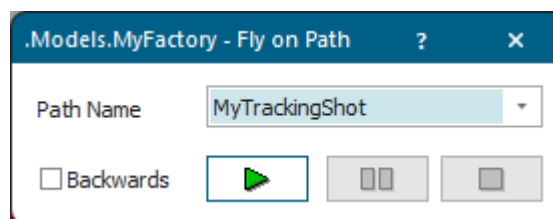
Note

You have to define a **camera animation path** on the **Tab Camera Animation** of the respective *Frame*.



To move the animation camera:

- Change to the **View** ribbon tab and click **Animate**.
- Select the name of the animation path for the selected object in the dialog **Fly on Path** or for the root object of the scene from the list **Path Name [MU animation]**.



- Select **Backwards** if you want the camera to move backward toward the starting point of the animation path. *Plant Simulation* automatically enters a negative velocity into the text box when you select the check box.
- To start the animation, click **Play** .

3D now shows the scene through the lens of the animation camera, which moves on the path you designated with a lens offset of (0, 0, 0.1).

- To pause the animation of the camera, independent of the simulation, click **Pause** .
- To stop the animation of the camera, independent of the simulation, click **Stop** .

At the end of the animation 3D sets the main camera to the last point where the animation camera was located and deletes the animation camera.

To test the camera animation path you defined, click the tab **Test** in the dialog **Edit Path**, select an **Animated graphic**, and start the test animation.

Compare [Work with Animation Paths](#)

Back to [Fly on a Defined Path Through the Scene](#)

Controlling Your View in the Scene

Control Your View

The *3D window* provides several ways of setting how you look at the scene, i.e., of controlling the camera that represents the eyes of the viewer looking at the scene.

You can:

- [Set the Main Directions](#)
- [Set View Points](#)
- [Attach a Camera to an Object/Frame and Detach it](#)
- [Animate the Camera](#)

Back to [Visualizing the Material Flow](#)

Set the Main Directions

Independent of the active camera setting, you can adjust the view to any main direction of the view or of an object at any time.

When you selected an object, the *3D window* zooms out this object and adapts the direction of the camera to the selected main direction of the coordinate system of the object.

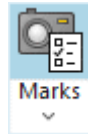
Compare [Align the View to the Main Directions](#)

Back to [Control Your View](#)

Set View Points

You can save any view of the scene, i.e., the position and orientation of the main camera within the scene, at any time, and return to this view later on.

You can also save the active camera setting, together with the active root object of the scene, with a



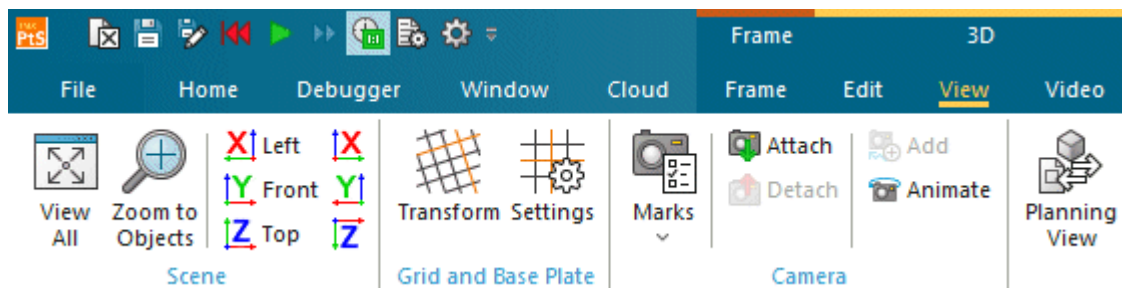
meaningful name as one of the **Camera Marks**.

As opposed to the view points that 3D saves automatically, you can save any number of **Camera Marks**, which it saves in the model file. You can then rename and delete them at will.

Back to [Control Your View](#)

Attach a Camera to an Object/Frame and Detach it

You can attach a camera to any object, so that you can view the scene through the lens of this object, and detach a camera from it again, when you do not need it any more. When the object moves through the scene, for example because you started the simulation, the camera will move along with the object you attached it to.



In addition, *Plant Simulation*:

- Visualizes the camera you attached, i.e., inserts an animatable object that represents the camera.
- Enables you to change between **object view** and **normal view**.
- Enables you to manually transform the camera you attached: By changing the view while in object view, or by manipulating the object, while in normal view.
- Enables you to set the speed and the direction of the animation.

- Enables you to pause, continue, and stop the camera animation independent of other animations and simulations.

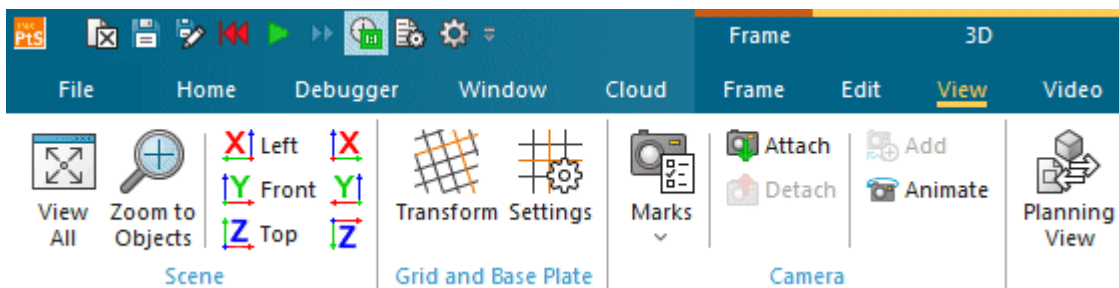
Compare [Attach a Camera to an Object](#)

Compare [Detach the Camera from an Object](#)

Back to [Control Your View](#)

Animate the Camera

You can make the camera move on any animation path of the selected object or of the root object of the scene. Here, you fly on this path and view the scene from it.



In addition, *Plant Simulation*:

- Visualizes the animated camera, i.e., inserts an animatable object that represents the camera.
- Enables you to change between **animation view** and **normal view** during the animation.
- Enables you to manually transform the animated camera: By changing the view while in animation view, or by manipulating the object, while in normal view.
- Enables you to define the speed of the animation and the direction of the animation.
- Enables you to pause, continue and stop the camera animation, independent of other animations and simulations.

Compare [Animate the Object Camera](#)

Compare [Fly on a Defined Path Through the Scene](#)

Back to [Control Your View](#)

Working with the Grid

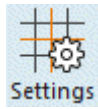
Work with the Grid

Plant Simulation uses the grid as the surface onto which it inserts objects.

Be aware that showing or hiding the grid affects how objects behave:

- When you **show** the grid, manipulating objects always refers to the **grid plane**.
- When you **hide** the grid, manipulating objects refers to your **view plane**, i.e., the plane perpendicular to the direction in which you look at the scene.

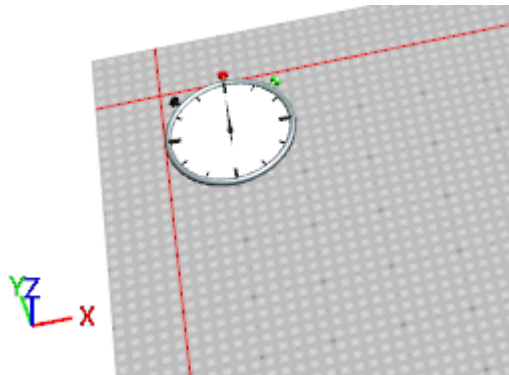
To change and define grid settings for the current simulation model and to open the dialog **Edit Grid and**



Base Plate Settings, click **Settings** on the **View** ribbon tab.

To change the grid settings for new models, select **File > Preferences > 3D [preferences]**.

The grid in the 3D window looks like this. The origin of the grid is located in the top left corner of the window offset by one grid unit to the right and down each.

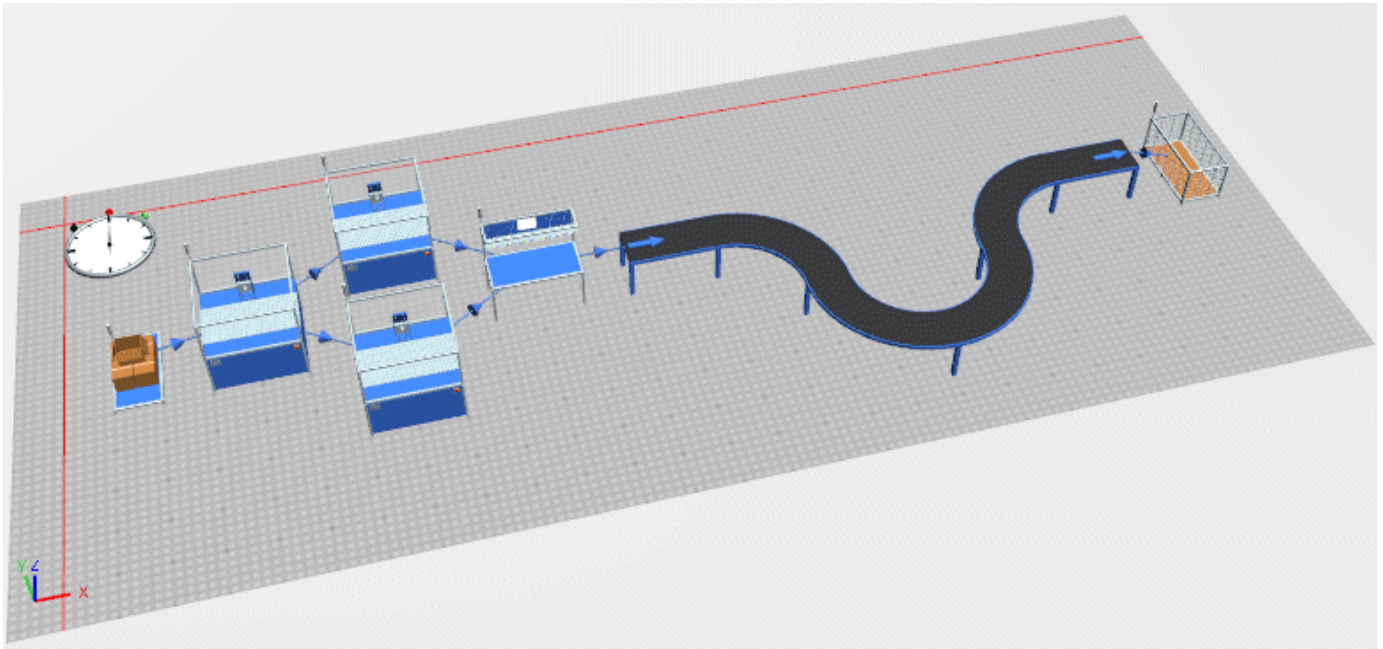


Plant Simulation re-creates the grid after each change you make in the model when it redraws the scene.

So, when you insert and move an object toward the borders of the grid, it automatically extends in that direction. The grid always expands or shrinks to show all objects in the scene.

As the grid is intended to be a modeling help when creating the model, you will usually hide it when presenting your model. To do so, press the **Ins** key on the keyboard.

Plant Simulation adjusts the dimensions and the borders of the grid to the position of the objects shown in the scene.



The origin of the grid, i. e., the lines that are red by default, always are part of the grid.

The grid size always is a multiple of the smallest visible grid distance.

You can:

- [Show and Hide the Grid](#)
- [Set Grid Properties](#)
- [Edit Grid Lines](#)
- [Position the Grid on Different Planes](#)
- [Move the Grid in the Window](#)

After you **inserted objects** into your simulation model, you can **align them to the grid** , **make them snap to the grid** , **make them snap to other objects** , or **arrange them on the grid**. To be able to do so, make sure that **Show Grid**  **Grid** is on.

Back to [Visualizing the Material Flow](#)

Show and Hide the Grid

You can show or hide the grid in the 3D window containing your simulation models.

To show or hide the grid, do one of the following:

- Click **Show Grid**  **Grid** on the **View Ribbon Tab**.
- Press the **Ins** key on your keyboard.

Compare [Insert Objects into the Scene](#)

Compare [Simultaneously Paste Multiple Copies](#)

Back to [Work with the Grid](#)

Set Grid Properties

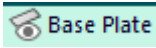
You can set the properties of the grid in the simulation model to meet your modeling needs.



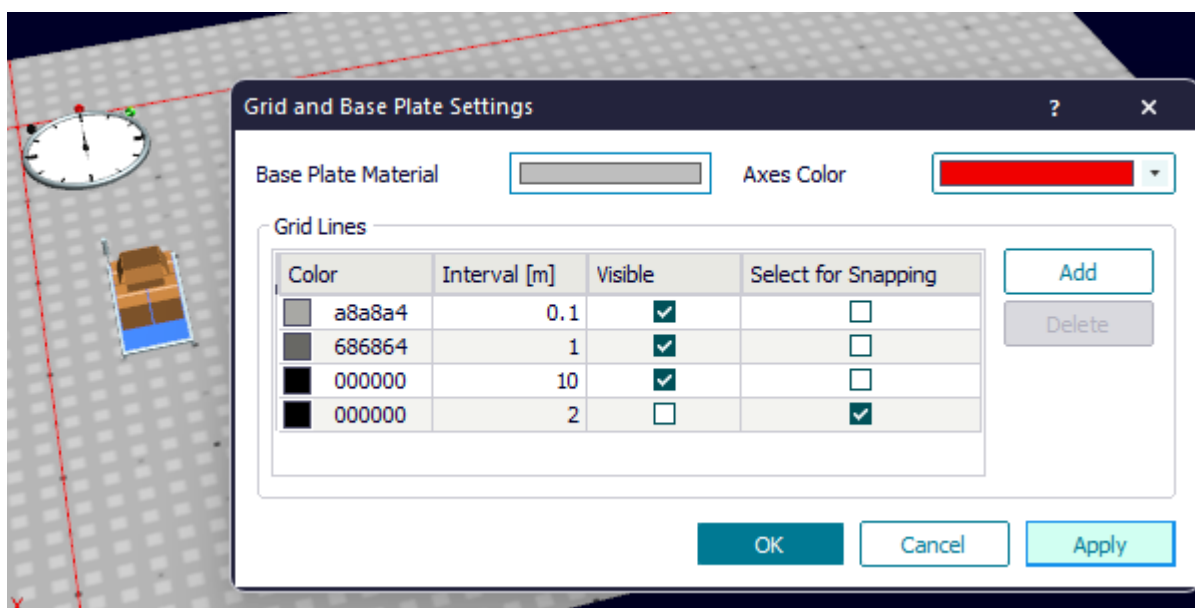
To do so, click **Edit Grid and Base Plate Settings**  on the **View** ribbon tab.

- To assign a material to the base plate below the grid lines in the *Frame*, click the color box. Then select individual components of the material in the dialog **Material** on the [Tab Material](#).

Note

To show or hide the **Base Plate**, click **Show Base Plate**  on the **View** ribbon tab.

- To assign a color to the coordinate axes, consisting of the x-axis and the y-axis, click the arrow in of the drop-down list **Axes Color** and select a color.
- To define the [settings](#) of the grid lines, edit them in the table on the tab.



Back to [Work with the Grid](#)

Edit Grid Lines

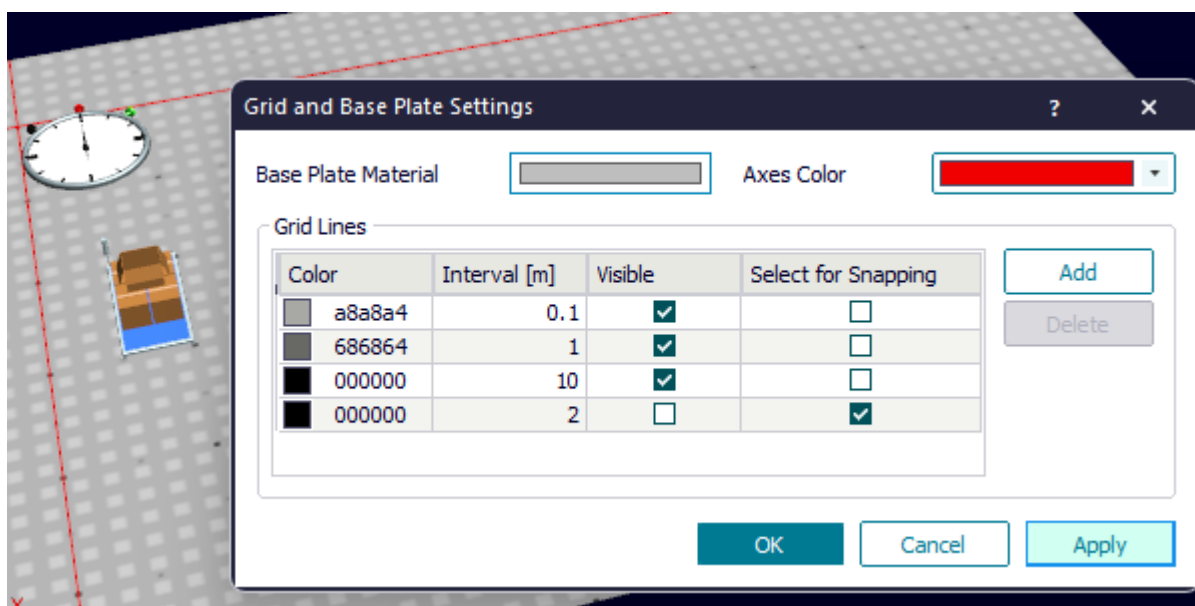
You can edit existing grid lines or add new grid lines.

You do this in the group box **Grid Lines**.

- To add another grid line, click **Add**.

Enter the distance between two grid lines, which you want to add; we entered 2 meters.

Select a color for the grid line, we selected a **bright blue**.



- To show this grid line in the grid in the scene, select **Visible**.
- Then, select if objects you insert **snap to this grid line**. Making objects snap to grid points enables you to precisely place objects with the mouse.
- To edit an existing grid line, select it in the list and edit the **distance** between the grid lines, their **color**, what objects, which you insert, **snap to**, and if this grid line is **visible** or not.

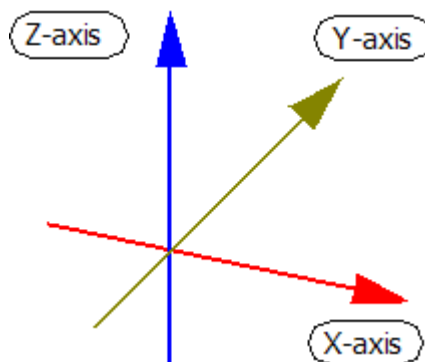
Back to [Work with the Grid](#)

Position the Grid on Different Planes

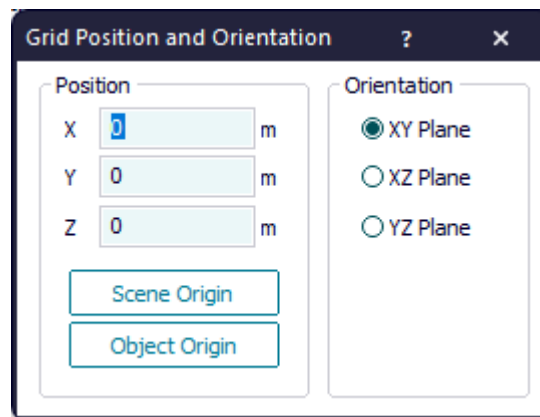
You can display the grid on the different planes in three-dimensional space.

As you remember *Plant Simulation* uses a left-handed coordinate system.

The **x-axis** runs from **left-to-right**, the **z-axis** runs from **bottom-to-top**, and the **y-axis** runs from **front-to-back**.



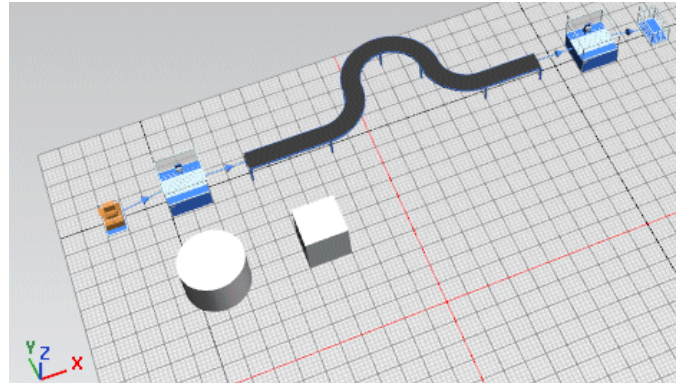
- To place the grid on the planes defined by the axes, click **Transform** on the **View** ribbon tab and select the respective radio button in the dialog **Grid Position and Orientation**.



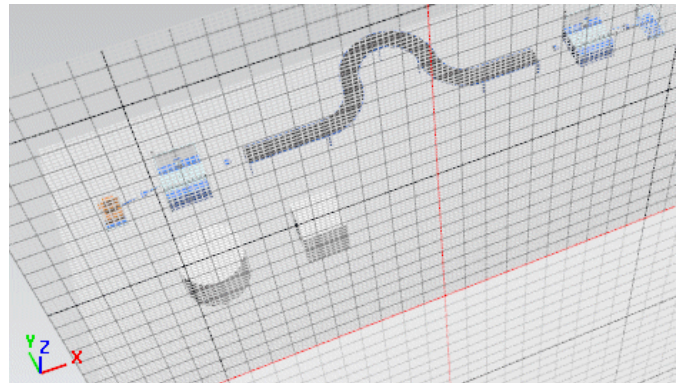
Note

Plant Simulation does not save the settings you select in this dialog. They only apply as long as the dialog is open.

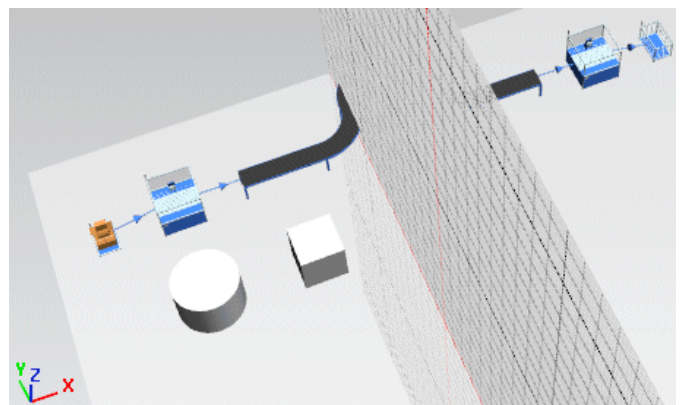
To position the grid on the plane defined by the x-axis and the y-axis, select **Orientation > XY Plane**.



To position the grid on the plane defined by the x-axis and the z-axis, select **Orientation > XZ Plane**.



To position the grid on the plane defined by the y-axis and the z-axis, select **Orientation > YZ Plane**.



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Move the Grid in the Window

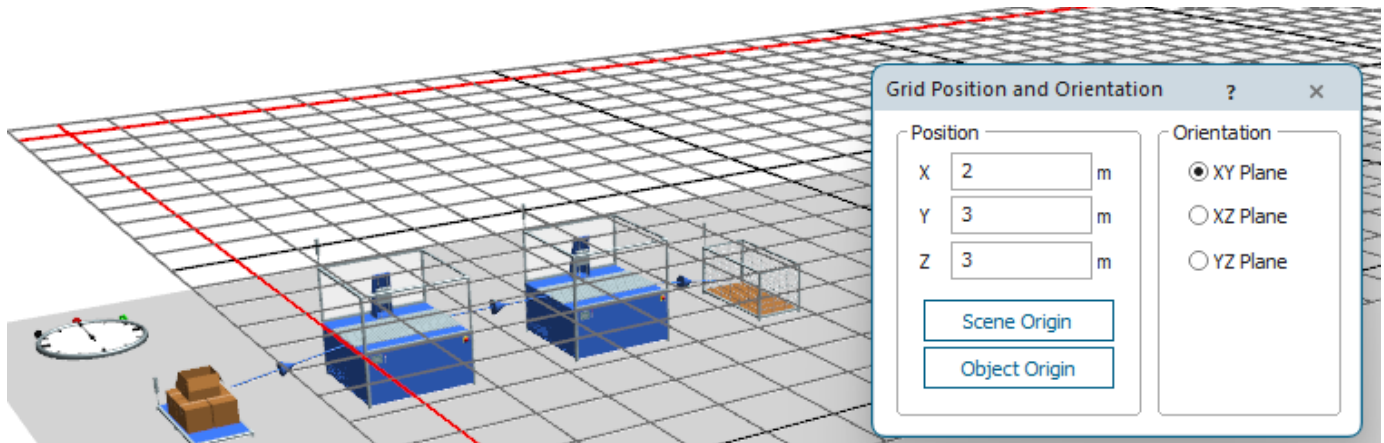
You can move the origin of the grid in the window of your simulation model.



To do so, click **Transform** on the **View** ribbon tab and roll the mouse wheel in the text boxes and click the buttons below **Position** in the dialog **Grid Position and Orientation**.

Note

3D does not save the settings you select in this dialog. They only apply as long as the dialog is open.

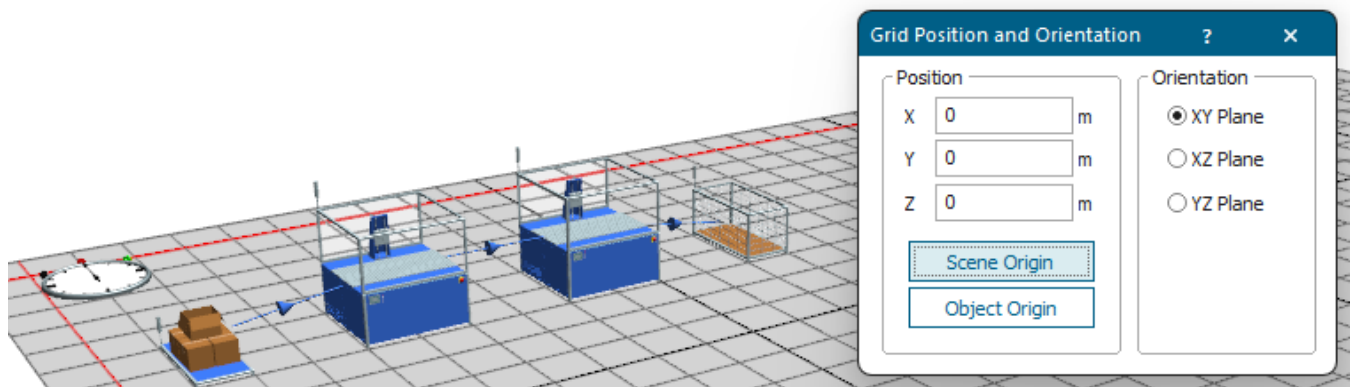


Note

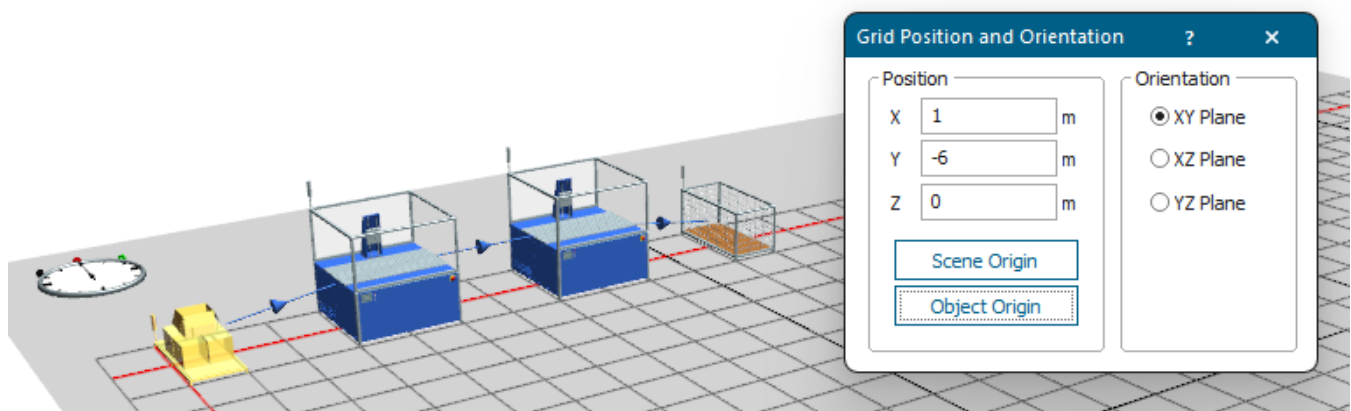
When moving the origin of the grid, *Plant Simulation* uses the length unit you selected under **Model Settings > Units > Length**.

- To move the position of the **Origin** of the grid on the **x-axis** by
 - 0.01** length units to the **right**, click in the text box, hold down **Ctrl**, and roll the mouse wheel forward.
 - 0.1** length units to the **right**, click in the text box and roll the mouse wheel forward.
 - 1** length unit to the **right**, click in the text box, hold down **Shift**, and roll the mouse wheel forward.
 - 0.01** length units to the **left**, click in the text box, hold down **Ctrl**, and roll the mouse wheel backward.
 - 0.1** length units to the **left**, click in the text box and roll the mouse wheel backward.
 - 1** length unit to the **left**, click in the text box, hold down **Shift**, and roll the mouse wheel backward.
- To move the position of the **Origin** of the grid on the **y-axis** by
 - 0.01** length units to the **back**, click in the text box, hold down **Ctrl**, and roll the mouse wheel forward.
 - 0.1** length units to the **back**, click in the text box and roll the mouse wheel forward.
 - 1** length unit to the **back**, click in the text box, hold down **Shift**, and roll the mouse wheel forward.
 - 0.01** length units to the **front**, click in the text box, hold down **Ctrl**, and roll the mouse wheel backward.
 - 0.1** length units to the **front**, click in the text box and roll the mouse wheel backward.

- 1 length unit to the **front**, click in the text box, hold down **Shift**, and roll the mouse wheel backward.
- To move the position of the **Origin** of the grid on the **z-axis** by
 - 0.01 length units **up**, click in the text box, hold down **Ctrl**, and roll the mouse wheel forward.
 - 0.1 length units to the **up**, click in the text box and roll the mouse wheel forward.
 - 1 length unit to the **up**, click in the text box, hold down **Shift**, and roll the mouse wheel forward.
 - 0.01 length units to the **down**, click in the text box, hold down **Ctrl**, and roll the mouse wheel backward.
 - 0.1 length units to the **down**, click in the text box and roll the mouse wheel backward.
 - 1 length unit to the **down**, click in the text box, hold down **Shift**, and roll the mouse wheel backward.
- To reset the values for the x-axis, the y-axis and the z-axis to their predefined default values of 0, 0, 0, click **Scene Origin**.



- To set the values for the x-axis, the y-axis and the z-axis to the values of the position of a single object you selected, click **Object Origin**.



Back to [Work with the Grid](#)